

STATEMENT OF PURPOSE

Applicant: Xuan Thanh Trinh

Program: Ph.D. in Electrical, Electronics and Computer Engineering

Proposed Advisor: Assistant Professor Van-Hai Bui

Institution: University of Michigan-Dearborn

1. Introduction

As a final-year Electrical Engineering student at Hanoi University of Science and Technology (HUST), my research at the Power Grid and Renewable Energy (PGRE) Laboratory focuses on developing distributed mathematical optimization models for decentralized energy management. While this foundation drove my graduation thesis on microgrid resilience, I recognized that classical optimization alone cannot scale to the real-time computational demands of modern power systems. To overcome these computational bottlenecks, I require advanced training in scalable, cyber-physical control systems. The Ph.D. program at UM-Dearborn perfectly aligns with this need, providing the rigorous framework I need to become a highly specialized research engineer capable of designing decentralized architectures for high renewable penetrations.

2. Academic Background

Ranking top 3 in HUST's Advanced Program in Electrical Engineering and Renewable Energy instilled in me a rigorous systems-level perspective. Rather than isolating individual grid components, I trained to deconstruct complex power networks into dynamic, tractable optimization problems. This multi dimensional mathematical foundation drives my independent research trajectory, allowing me to architect solutions that bridge abstract algorithms with real-world physical constraints.

Safeguarding legacy power systems against cascading failures requires a paradigm shift from rigid, top-down architectures to decentralized microgrid communities. Recognizing this necessity, my research at the PGRE lab focused on subdividing large-scale distribution networks into autonomous microgrids empowered by peer-to-peer energy trading.

A reliable and robust energy market must guarantee physical reliability alongside economic fairness. To establish and validate this framework, I integrated rigorous grid physics (AC-OPF) with distributed optimization (ADMM) across both radial and ring-connected topologies. This architecture drastically reduced operational costs while strictly bounding voltage levels within the critical $[0.95, 1.05]$ pu threshold, ensuring community-level trading never compromises the wider public grid.

Furthermore, modern grid architectures must dynamically shield communities from severe local failures. To achieve this fault tolerance, I layered Model Predictive Control (MPC) over the trading baseline to coordinate emergency power exchanges and backup diesel generation. During simulated localized outages, this temporal coordination empowered neighboring networks to assist a failing grid, successfully reducing forced load shedding by 80% and ensuring uninterrupted power for essential civilian infrastructure.

However, the computational latency of non-linear physical constraints remains a critical barrier. While my mathematical models guaranteed strict physical safety, the overhead of re-optimizing AC-OPF equations rendered classical solvers unscalable for standard tertiary dispatch intervals. As power grids integrate highly volatile renewable sources, this computational bottleneck threatens widespread stability. Overcoming this limitation through learning-based AI control is a societal imperative, directly motivating my decision to pursue a Ph.D.

3. Research Interests

To resolve the trade-off between computational tractability and physical grid security, my Ph.D. research will focus on architecting Physics-Informed AI frameworks for real-time power system dispatch. While pure data-driven models accelerate computational speeds, their deployment frequently fails because unconstrained black-box policies inherently violate non-linear AC power flow equations and thermal limits. I intend to

synthesize these paradigms by deploying my deterministic optimization models in a dual capacity: as expert benchmarks to generate mathematically rigorous optimal dispatch datasets, and as differentiable safety filters—leveraging convex relaxations like SOCP—that project AI-generated setpoints into a verifiable feasible region. By incorporating structural grid constraints into both the offline training and online execution agents, I aim to pioneer certifiably safe and physically resilient dispatch architectures.

4. Academic and Research Alignment

Realizing that architecting Physics-Informed AI frameworks requires an ecosystem that seamlessly bridges advanced machine learning with rigorous mathematical optimization, I found the Ph.D. program at UM-Dearborn to be the perfect catalyst for my research. To construct mathematically guaranteed safety filters for these AI agents, I require a stringent theoretical foundation. The interdisciplinary scope of the EECE curriculum will systematically bridge my theoretical gaps, providing the comprehensive mathematical and computational toolkit necessary for this architecture through advanced courses such as IMSE 505 (Optimization) and ECE 5001 (Analytic and Computational Math). Furthermore, the robust computational facilities at UM-Dearborn, paired with its strategic ties to regional energy sectors, provide the exact environment I need to rigorously test and validate these theoretical architectures against real-world grid scenarios.

Within this dynamic ecosystem, my research trajectory strongly aligns with the ongoing work of Prof. Van-Hai Bui in autonomous energy markets. His recent publications on FairMarket-RL and XRL-LLM investigate the integration of Large Language Models into multi-agent energy trading, an area highly relevant to my interest in learning-based policies. By joining his laboratory, my foundation in mathematical optimization will allow me to seamlessly support his ongoing research while advancing my own theoretical frameworks. Crucially, Prof. Bui's mentorship and profound expertise in AI surrogates will provide the exact guidance I need to successfully integrate deterministic AC safety filters into learning-based policies, ultimately realizing my vision of a physically resilient smart grid.

5. Career Objectives

Following the completion of my Ph.D, I intend to deepen my expertise as an R&D scientist within the U.S. energy sector before ultimately returning to Vietnam to contribute to the technical modernization of our national grid infrastructure. By repatriating advanced learning-based control methodologies, my long-term vision is to help transition our power architectures toward autonomous, resilient networks capable of stabilizing the rapid integration of clean energy sources. Ultimately, the Ph.D. program at the University of Michigan-Dearborn provides the exact interdisciplinary ecosystem and mentorship necessary to launch this career trajectory.